

3.3 Integrated Development Environment (IDE)

آئی ڈی ای (IDE) ایک ایسا سافٹ ویئر ہے جس میں کوڈ لکھنا، چلانا اور غلطیاں درست کرنا ایک ہی جگہ ممکن ہوتا ہے۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 04 of 24

3.3 Integrated Development Environment (IDE)

LEARN IT

An **IDE (Integrated Development Environment)** is a software tool that helps programmers **write, run and manage code in one place**. It is a complete workspace for programming. "Integrated" means many tools are joined together in one window.

Key features of an IDE:

- **Code Editor:** where you write and edit code.
- **Syntax Highlighting:** colours different parts of the code (keywords, strings, numbers) to make it easier to read.
- **Auto-completion:** suggests code while you type (type `pri` and it offers `print`).
- **Run/Execute Tool:** runs the program with one click.
- **Debugger:** helps you find and fix errors, for example by pausing the program at a chosen line.

WORKED EXAMPLE

Without vs with an IDE. Without an IDE you would write code in Notepad, save it, open a terminal and type `python file.py`, then go back to Notepad to fix mistakes. With an IDE you write, run (press the ► Run button) and debug in the same window, and errors are underlined while you type.

REMEMBER IT

IDE features — **"E-S-A-R-D"**: Editor, Syntax highlighting, Auto-completion, Run, Debugger.

COMMON MISTAKE

An IDE does not write the program for you and does not catch logical mistakes. It only makes writing, running and debugging easier.

اردو میں سمجھیں

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CHECK YOURSELF

1. What does "integrated" mean in IDE?
2. Which IDE feature suggests code while you type?

Answers: 1. Many programming tools (editor, runner, debugger) are combined in one program. 2. Auto-completion.

Next lecture: 3.3.1 Visual Studio Code (VS Code)